

Let's Encrypt HTTPS — Integration Status

Revision: 1 **Last modified:** 2026-07-01T12:15:00Z **Status:** In progress — Phase 1 cert-analyzer landed; issuance/renewal phases pending. Production cutover OPERATOR-BLOCKED (design §9 real-domain gate). **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc for the Let's Encrypt HTTPS workstream (task #59). **Companion:** design ../LETSENCRYPT_HTTPS.md · plan ../LETSENCRYPT_HTTPS_PLAN.md · summary Status_Summary.md

Operator-blocked items (read first — §11.4.45 O(1) surface)

Item	Why blocked	Unblock condition
Phase 4 — LE staging end-to-end (real DNS-01)	Needs a real DNS provider + an API token stored as a Podman secret (§11.4.10 — value never in git/.env). Operator chose <i>defer real domain</i> (Phase 0).	Operator provides the DNS provider name + a scoped API token, and names a test hostname.
Phase 6 — production cutover	Real public domain, real DNS-01 for that domain, single operator-gated go-live (design §9).	Operator authorises the go-live with the real domain + token + host firewall opening.

Operator decisions captured (Phase 0 — §11.4.66)

Decision	Choice	Date
TLS client / terminator	Caddy auto-HTTPS (in-process ACME client + terminator; no cron/CI/hook renewal — §11.4.156)	2026-07-01

Decision	Choice	Date
ACME challenge type	DNS-01 (works without inbound :80/:443 to the host)	2026-07-01
Rollout order	Hermetic (Pebble) + LE-staging first; defer real domain — prod is one later operator-gated cutover	2026-07-01

Phase status (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Validation	Evidence
0 — Operator decisions + scaffolding	Capture the 3 design §9 decisions; scaffold the plan	PASS	decisions table above; committed LETSENCRYPT_HTTPS.md + _PLAN.md @ fddd25c
1 — Cert-analyzer (client/challenge-agnostic)	Pure offline analyzer over a resulting PEM (cert_not_expired / cert_days_remaining / cert_san_matches / cert_chain_roots_in / cert_renewal_due) + golden-good/golden-bad self-validation (§11.4.107(10))	PASS	selftest 37/37; guard GREEN+RED PASS; §1.1 mutation (SAN exact→substring) → guard FAIL, restore byte-identical md5 a06e0f89...; qa-results/letsencrypt/cert-analyzer/ + qa-results/regression/cert_analyzer_selfvalidation/
1 — Custom Caddy image w/ DNS-01 provider module	Build a Caddy image bundling the chosen DNS provider plugin (via the containers submodule, rootless — §11.4.76/.161)	PENDING	—
2 — Compose service + Podman secret + cert volume	Wire Caddy service + secret NAME/PATH (no value) + persistent cert volume; no real issuance yet	PENDING	—
3 — Hermetic local-ACME with	Caddy issues a real cert against a local Pebble ACME server, fully offline/hermetic; cert-	PENDING	—

Phase	Scope	Validation	Evidence
Pebble (anti-bluff core)	analyzer asserts the result		
4 — LE staging end-to-end (real DNS-01)	Real LE staging endpoint + real DNS-01	OPERATOR- BLOCKED	needs DNS provider + token (§11.4.10)
5 — Renewal + rotation simulation (zero- downtime)	Force near-expiry → observe Caddy renew → analyzer confirms new cert; no downtime	PENDING	—
6 — Production cutover + rollback runbook	Real domain go-live	OPERATOR- BLOCKED	operator go-live gate (design §9)

Honest boundary (§11.4.6)

Only Phase 0 and the Phase-1 **cert-analyzer** are DONE with captured evidence. The analyzer validates a resulting certificate regardless of how it was obtained, so it is ready to assert the output of Phases 3–6 the moment those land. No issuance, renewal, or rotation path has been exercised yet — those rows are honestly PENDING / OPERATOR-BLOCKED, never a metadata-only PASS (§11.4 / §11.4.1). The hermetic Pebble path (Phase 3) is the next autonomous target; the real-domain paths (4, 6) wait on the operator gate.